

Peer Influence Effects on Risk-Taking Decision Making During a BART Task

Anna Herlache, Jacob Folkers, Lydia Ellis, Mairin Syth

Department of Psychology, University of Wisconsin-Madison

Psych 505: Neurobiology of Learning and Decision-Making

Group 2, section 306

5/3/2026

Abstract (By: Lydia)

Social contexts influence behavior in many ways: adolescents make riskier choices when with peers, reward neurocircuitry activates stronger in social environments, and people make decisions according to social dynamics. Past research about peer influence and risk-related behavior has focused on adolescents, but college students also experience highly social environments of many different sizes. The present study evaluated how peer presence and group size influence risky decision-making during a BART-style task. We hypothesized social facilitation and reward sensitivity would increase risk-taking in small group settings versus individual trials, but that this effect wouldn't scale linearly in large groups. Participants completed a questionnaire self-assessing their proclivity for risk-taking behavior before completing trials alone, then in groups of three and six. Our sample of 37 participants showed people took more risks when alone, not in groups. These findings suggest that social context shifts how individuals weigh reward vs. potential loss, rather than simply increasing risk-taking. Risk-taking changed across trials, showing that participants calculate risk according to early trial outcomes and implicates dopamine-driven learning processes.

Introduction (By Anna)

Making decisions involving risk is a fundamental aspect of everyday life. Individuals routinely make choices involving uncertainty, potential rewards, and potential negative outcomes. These decisions are made using interacting neural and neurochemical systems in the brain that encode subjective value, risk, and uncertainty. Neuroimaging and neurophysiological research has identified key roles for brain regions such as the striatum, prefrontal cortex, and anterior insula in representing reward, cognitive control, and risk evaluation, respectively¹. The striatum is heavily modulated by dopaminergic signaling and is particularly involved in reward processing and motivation while the prefrontal cortex is involved in executive control and the regulation of behavior in the face of risk. The insula has been found to trace uncertainty and subjective risk, contributing to emotional responses during decision making. Additionally, neurotransmitters such as dopamine and serotonin play a critical role in modulating these decision-making processes and influence how individuals balance potential rewards and possible losses². Together, these brain regions and systems form a neural basis for evaluating and responding to decisions involving risk.

Social environments can exert a powerful influence on decision-making, especially in contexts involving risk. There is a substantial body of research demonstrating that peer presence increases risk-taking behavior in adolescence, yet much less is known about

how social influence shapes risky-decision making in adults. Social contexts can alter how individuals weigh costs and benefits, and research in adolescents has shown peer presence to enhance activity in reward-related neural systems, such as the striatum, and promote riskier choices^{3,4}. This effect can be explained by social facilitation, where the presence of others amplifies dominant responses, frequently toward reward-seeking in social settings. Even in the absence of explicit pressure, group norms and social expectations can shift individual preferences, leading individuals to align their behavior with others in ways that could either increase or decrease risk⁵.

In contrast to the extensive literature on adolescents, the mechanisms underlying peer influence on adult risk-taking behavior are much less understood, particularly with respect to social environments and features of the social environment such as group size. According to Social Impact Theory, the influence of a social environment is a function of the strength, immediacy, and number of others present, with effects that follow a non-linear trend, suggesting that adding more peers eventually leads to a plateau in behavioral change⁶. Additionally, social influence does not uniformly increase risk-taking in adults. Some social contexts may promote more cautious behavior compared to risk, particularly when individuals anticipate evaluation or seek to avoid negative judgement. College-aged individuals may be especially sensitive to social environments in the face of risk, as they are more mature and independent than adolescents but are still in highly social environments and are exposed to heightened evaluation and comparison.

The present study aims to fill the gap in current research by investigating how peer presence and group size influence risk-taking behaviors in college-aged adults using methods similar to the Balloon Analogue Risk Task (BART)⁷. By comparing baseline risk-taking in individuals to decision-making across social contexts that vary in both the presence and number of peers, this study examines how social context and facilitation shape risky-decision making in adults.

We hypothesize that social facilitation and reward sensitivity will significantly influence risk-taking behavior on the BART in the presence of peers. Specifically, we expect participants to show a noticeable shift toward increased risk-taking from the individual condition to the small group condition (group of three), but that this effect will not increase linearly with group size. In accordance with Social Impact Theory, we predict a plateau in social influence, where the transition from a small group to a large group (group of five) results in diminishing marginal returns in behavioral change. Furthermore, we predict that individual baseline risk tendencies will moderate these effects: risk-prone individuals will exhibit a facilitated increase in risk-taking across all group conditions, while risk-averse individuals will be more sensitive to social monitoring

and group norms, either increasing their risk-taking to align with peers or maintaining caution to avoid negative social evaluation. Overall, we expect peer presence and group size to interact with individual differences to shape risky decision-making through the mechanisms of conformity and evaluation apprehension.

Methods (By: Jacob)

37 undergraduate students from the University of Wisconsin–Madison participated in this study. All participants completed the study in a single session using a within-subjects design in which each individual experienced three conditions that varied by social context: an individual condition, a small-group condition (group of three), and a larger-group condition (group of six).. The order of conditions was kept consistent across participants.

Before beginning the behavioral task, participants completed a brief self-report questionnaire assessing their perceived tendency to engage in risky behavior. This measure was used to examine how participants' subjective perceptions of risk compared to their actual behavior during the task.

Participants then completed a computerized task modeled after the Balloon Analogue Risk Task (BART). In each trial, participants were presented with a virtual balloon and instructed to press a button to inflate, or “pump,” the balloon. Each pump increased the potential reward associated with that trial but also increased the probability that the balloon would explode. If the balloon exploded, participants lost all accumulated reward for that trial. Alternatively, participants could choose to stop pumping at any point and “cash out,” securing the reward earned up to that point. This task required participants to continuously weigh the trade-off between increasing reward and the risk of loss, providing a behavioral measure of risk-taking.

Social context was manipulated by varying the presence and number of peers during the task. In the individual condition, participants completed the task alone. In the small-group condition, participants completed the task while exposed to a group of three peers. In the larger-group condition, participants completed the task in the presence of six peers. These conditions were designed to test how peer presence and group size influence risk-taking behavior.

The primary dependent variable was average adjusted pumps, defined as the mean number of pumps on trials in which the balloon did not explode, which is a standard index of risk-taking in BART paradigms. Balloon explosion rate was also examined as a secondary measure of risk-taking behavior. Self-report risk scores were compared with behavioral performance to assess the relationship between perceived and actual risk-taking. In addition, changes in behavior across trials were examined to evaluate

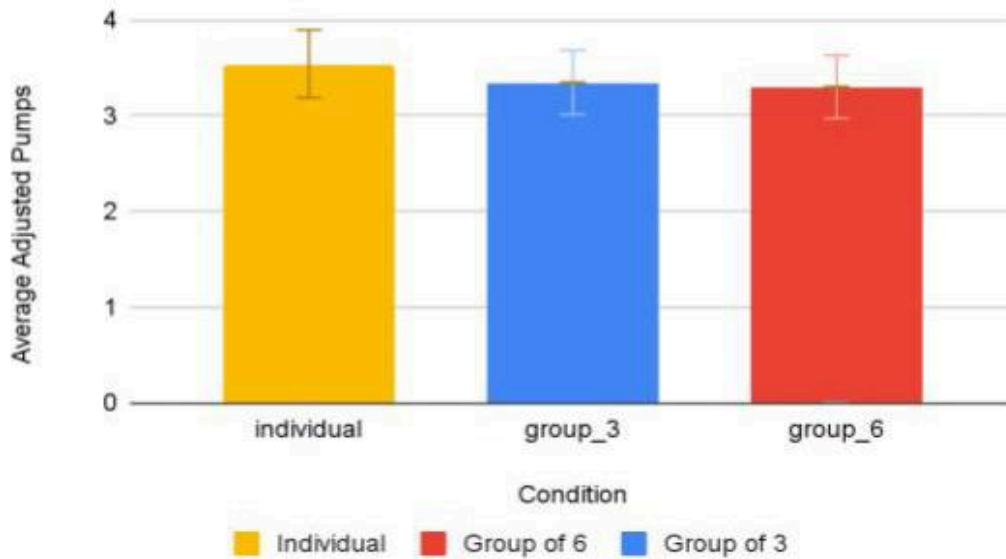
potential learning effects, such as adjustments following early successes or losses. Paired-sample t-tests were conducted to compare risk-taking behavior across the three conditions and determine whether social context significantly influenced decision-making.

Results (By: Mairin, Anna)

In the self-report questionnaire, there was a general positive relationship between self-reported risk-taking behaviors and their actual risk-taking behaviors. Participants who rated themselves as higher risk tended to show higher risk taking behaviors. However, this was not consistent across all participants, as some showed higher risk-taking behaviors despite moderate self-ratings. Results from the BART task were not consistent with the original hypothesis. Based on previous behavior patterns demonstrated in the literature, participants were expected to demonstrate riskier behavior patterns in the group conditions. Nevertheless, results showed that participants were more risk seeking in the individual condition compared to both small and large group conditions. While there were no significant differences in average adjusted pumps between conditions, participants showed significantly higher balloon pop rates in the individual condition. Additionally, risk-taking behavior changed across trials. Based on the average patterns shown throughout trials, participants typically exhibited a slight dip in risky behavior around trial 2 and a peak around trial 4, which suggests that participants may have initially acted cautiously and adjusted their behaviors based on early outcomes.

Figure 1

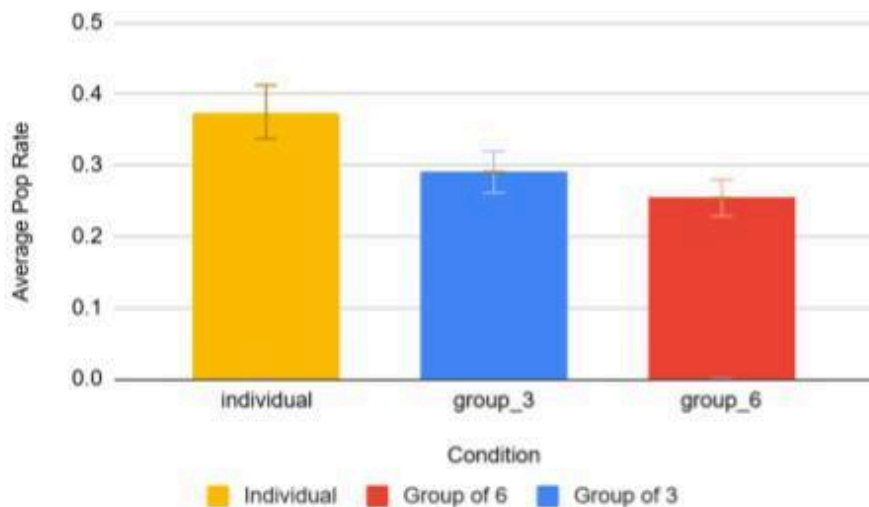
Average Risk Behavior (Adjusted Pumps) Between Conditions



Note. Figure 1 depicts average adjusted pumps of participants between individual (Mean= 3.539), groups of 3 (Mean= 3.345), and groups of 6 (Mean=3.299) conditions.

Figure 2

Average Pop Rate Between Conditions



Note. Figure 2 depicts average pop rates of participants between individual (Mean= 0.375), groups of 3 (Mean= 0.290), and groups of 6 (Mean=0.254) conditions.

Table 1

Average Behavioral Risk Rating Between Conditions

Condition	Mean (Average)	Standard Deviation
Individual	3.444	1.004
Group of 3	3.235	0.919
Group of 6	3.203	0.925

Note. Table 1 depicts average behavioral risk between conditions of individual, group of 3, and group of 6 trial conditions.

Table 2

Behavioral Risk Rating Between Conditions

P-values from paired-samples t-tests

Individual vs. Group of 3	$p = 0.137$
Individual vs. Group of 6	$p = 0.833$
Group of 3 vs. Group of 6	$p = 0.138$

Note. Table 2 depicts p-values from paired-samples t-tests comparing behavioral risk ratings between each trial.

A series of paired-sample t-tests were conducted to analyze behavioral risk across the three conditions. There was no statistically significant difference in adjusted pumps

between the Individual condition ($M=3.498$, $SD=1.004$) and the Group of 3 condition ($M=3.297$, $SD=0.942$), $t(36)=0.137$, $p > 0.05$. Similarly, there was no statistically significant difference between the Individual condition ($M=3.498$, $SD=1.004$) and the Group of 6 condition ($M=3.203$, $SD=0.925$), $t(36)=0.833$, $p > 0.05$. Lastly, there was no statistically significant difference in behavioral risk between the Group of 3 ($M=3.297$, $SD=0.942$) and Group of 6 ($M=3.203$, $SD=0.925$) conditions, $t(36)=0.138$, $p > 0.05$. With no statistically significant difference between the groups, we cannot conclude that peer presence or group size had an effect on decreasing or increasing risky behavior, compared to our hypothesis that the social environment would decrease risky decision-making behavior.

Conclusion (By: Lydia)

The present study examined how peer presence and group size influence risk-taking behavior in college-aged adults using a BART-style task. Contrary to our original hypothesis, participants demonstrated greater risk-taking behavior when alone compared to when they were in social contexts. Although differences in average adjusted pumps were not statistically significant, higher balloon explosion rates in the individual condition suggest that participants engaged in riskier decision-making in the absence of peers. These findings challenge the commonly held assumption, largely based on adolescent research, that peer presence uniformly increases risk-taking behavior.

One interpretation of these findings is that social contexts in adulthood may engage neural systems associated with cognitive control and social evaluation, leading to more regulated behavior. While prior work has shown that peer presence can enhance activity in reward-related regions such as the striatum, particularly in adolescents, our results suggest that in adults, the presence of others may instead increase engagement of the prefrontal cortex, which supports behavioral regulation and inhibition. Additionally, the anterior insula, which is involved in processing uncertainty and potential loss, may become more active in social contexts where individuals anticipate evaluation or judgment from others. Together, this shift in neural engagement could lead individuals to weigh potential risks more cautiously in group settings.

Changes in behavior across trials further suggest the involvement of reinforcement learning processes. Participants initially showed more cautious behavior, followed by increases in risk-taking, indicating that early outcomes influenced subsequent decisions. This pattern is consistent with dopamine-mediated reward prediction error signaling in the striatum, where unexpected outcomes shape future behavior. These findings highlight that risk-taking is dynamically influenced by both experience and social context, rather than being a fixed trait.

Several limitations should be considered. The relatively small sample size may have limited the ability to detect statistically significant differences across conditions. Additionally, the social manipulation may not fully capture real-world peer influence, as experimental settings lack the complexity and emotional salience of natural social environments. Future research should incorporate more ecologically valid designs and examine individual differences in sensitivity to social evaluation, which may influence how neural systems respond to peer presence.

The present study could also be extended to measure brain activity during the BART task to allow further insight into the neural underpinnings driving any difference in decision making behavior between the individual and group trials. The prefrontal cortex is one brain region with significant differences in grey and white matter volume as it develops from adolescence into adulthood⁸. These structural differences translate into behavioral differences as shown in research with mice whose performance during PFC-dependent tasks was markedly worse than adult mice, but performed similarly to adults during non-PFC-dependent tasks⁹. This makes the PFC especially relevant to our investigation of decision making in college-aged young adults in contrast to adolescent populations which are historically centered in the literature regarding risky decision making and peer influence. Utilizing functional near-infrared spectroscopy (fNIRS) methodology we could observe activity of the PFC across all experimental conditions of the BART task. fNIRS detects blood-oxygen-level-dependent (BOLD) signals similar to an MRI, but can be worn similar to an EEG cap allowing brain activity to be measured in real-world social settings. Tsunashima and Yanagisawa¹⁰ used fNIRS technology to measure brain activity of participants during a driving task and found the fNIRS measurements and MRI measurements showed the same data, showing the reliability of the newer fNIRS technology when compared to the more commonly used MRI. Expanding on our initial hypothesis we could expect to see decreased BOLD activity in the PFC during small group trials when participants are hypothesized to increase their risk taking behavior compared to individual trials, thus reducing recruitment of PFC areas, but if we hypothesize this effect will not scale linearly with the larger group size condition we would expect increased BOLD activity in the PFC indicating increased caution and decreased risk-taking behavior.

Overall, these findings suggest that peer influence on risk-taking in adults is more nuanced than previously assumed. Rather than simply increasing risk, social contexts may shift the balance between reward-related and control-related neural systems, resulting in more cautious decision-making. This underscores the importance of considering both developmental stage and underlying neural mechanisms when studying how social environments shape behavior.

Contributor Block

Programming: Jacob Folkers

Presentation: All members, Jacob Folkers (slide design and formatting)

Introduction: Anna Herlache

Methods:

Results & Discussion: Mairin Syth (Primary), Anna Herlache (figures and analysis)

Conclusion: Lydia Ellis

References: All members

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